

THE ART OF WAR: HOW TO SOLVE TRUE/FALSE QUESTIONS IN REAL ANALYSIS

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INTRODUCTION

Professor Clément Hongler was responsible for perhaps the most infamously difficult examination in recent memory in Analysis 1, when he set a 2018 exam containing 19 true/false questions, of which it sufficed to solve only 6 completely in order to pass the exam. In retrospect, however, one finds that the questions themselves are not too difficult, and the true challenge lies in being able to provide concise and accurate proofs, and particularly counterexamples when necessary. This document is intended to be a short guide on how to tackle these types of questions, for which many students are ill-prepared because they do not form a major part of their revision material, not often appearing in the exercise sheets or that vaunted Bible of real analysis, the Douchet.

Finally, I would like to impress upon the reader that this should be not considered some failsafe algorithm to tackle this type of question - this is but a sketch of useful tips and tricks that can go a long way towards honing your abilities. However, the only way to improve your mathematical intuition is through practice, practice, practice - do as many problems as you can!

If you spot any errors in this manuscript, then please let me know at

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or better still, in person if you run into me. Other feedback and suggestions for improvement are always welcome!

1 WHAT EXACTLY IS A TRUE/FALSE QUESTION?

As any good hunters would do, before setting off after our prey, we ought to understand exactly what it is we are dealing with. Luckily, it is actually quite easy to cut open and look at the anatomy of these wild beasts.

1.1 A little dissection

A true/false question can usually be split into three constituent parts:

- The *subject*, or the type of mathematical object we are dealing with. This is often one or more sequences, series, functions, sequences of functions... all the classic structures you have seen in the course.
- The *properties*, or the starting conditions we are supposing the subject satisfies.
- The *conclusion*, a mathematical statement usually depending on the subject and its properties, that is either always true as a consequence of them, or not always so.

A question can then be phrased as:

Given a **subject** satisfying **properties**, then we have that the **conclusion** holds.
Is this true or false? If true, then justify; if false then provide a counterexample.

From a logical standpoint, we are asking if a set of axioms $\mathcal{A}$ necessarily implies some consequence $\mathcal{B}$. A counterexample is therefore an example of some subject with the properties that does **not** imply the conclusion.

1.2 Is it true or is it false?

So the critical question is which of the two is correct, because before you can find a proof or a counterexample you need to know which one you should be looking for. Simply put, there is no easy way to know which one you should target: you should simply follow your gut instinct. Often, the question is designed to be counterintuitive; either the properties seem too weak to always imply the conclusion or the conclusion seems obvious and a counterexample unimaginable.

Nevertheless, you often find that going down one path leads to you either to what you were searching for or the ingredients for the other path. In other words, if you start looking for a proof that the properties imply the conclusion you might start eliminating subjects one by one until you stumble upon the very subject that is the counterexample. An ideal way to go about this is to consider additional properties (for example, what if a sequence is strictly positive) and then for this property, you either find a counterexample or a proof, allowing you to either conclude or note that a counterexample must not obey this property (so you can stop considering strictly positive sequences for your counterexample).

Alternatively, trying to find the counterexample might lead you to find a sequence of characteristics that it must possess as a consequence of the initial properties, and eventually you stumble onto the conclusion itself. To better illustrate this, let us consider a classic example of a true/false question.

Problem. Let $\{x_i\}_{i=0}^{\infty}$ be a sequence of real numbers such that $\sum_{i=0}^{\infty} x_i$ converges. Then $\sum_{i=0}^{\infty} x_i^3$ converges. True or false?

What does our intuition say about the behaviour of the sequence of cubes? Well, since the original sequence must tend to 0 (a necessary condition for the series to converge) it follows that the cubes of the terms are eventually contained between the terms themselves and 0, and so in a behaviour similar to the two gendarmes theorem, we would expect the sum to go towards 0. So let's set about trying to prove it.

Firstly, for strictly positive sequences, this is indeed clearly true - since the cubes are eventually bounded from above by the original terms, we can find an upper bound on the the sum of the cubes. Proving this rigorously is left as an exercise to the reader.

Now what happens if the sequence alternates between positive and negative? (Henceforth, alternating will mean every term, so constantly switching from positive to negative and back again). We can try repeating the trick of using the original terms to bound the sum of the cubes. Since we know the terms alternate between positive and negative, denote the sequence by $y_0, -y_1, y_2, -y_3, \dots$. We then have¹

$$\begin{aligned} \sum_{i=0}^{\infty} x_i^3 &= \sum_{j=0}^{\infty} y_{2j}^3 - y_{2j+1}^3 \\ &= \sum_{j=0}^{\infty} \underbrace{(y_{2j} - y_{2j+1})}_{\text{converges}} \underbrace{(y_{2j}^2 + y_{2j}y_{2j+1} + y_{2j+1}^2)}_{\rightarrow 0, \text{ positive}} \end{aligned}$$

This is actually somewhat inconclusive, because just because we multiplied individual terms by something that goes towards 0 does not allow us to conclude immediately². Nevertheless, if you test a few cases you will see that if $|x_i| = y_i$ is strictly decreasing then it will indeed converge. Now, it was incorrectly stated in the revision seminar that we were finished with this case at this point and that was what led us to consider sequences which do not alternate regularly but instead have two negative terms for every positive one (and this makes finding the counterexample rather easy) but as you can see, we didn't indeed conclude the alternating case properly. Indeed, we can even find a counterexample that is alternating after all, so to diverge a bit from the content of the seminar, let us find this other one instead.

As we mentioned, if y_i is strictly decreasing then it is not a counterexample. So what if it is not? Let us take then a sequence such that $y_0 > y_1, y_2 < y_3, y_4 > y_5, y_6 < y_7$, etc. Now, a condition for the original series of x_i to converge is just that the series of $y_{2i} - y_{2i+1}$ converges. Since the latter is now alternating as well, a sufficient condition for convergence is if $|y_{2i} - y_{2i+1}|$ goes to 0 and is strictly decreasing (you have seen this in the course; if a sequence is alternating and its absolute value goes monotonically to 0 then the resultant series converges, e.g. the alternating harmonic series). So let us pose $y_0 - y_1 = \alpha_0 > 0$ and $y_3 - y_2 = \beta_0 > 0$, and do a little algebra. If this seems a bit confusing, it will be much clearer soon. Firstly note that

$$x_0 + x_1 + x_2 + x_3 = y_0 - y_1 + y_2 - y_3 = \alpha - \beta$$

¹ Note that this trick is a "sufficient but not necessary condition" : if the sum when you group terms together diverges, then the original sum diverges. However, if the sum converges, this does not tell you anything about the original sum: the classic example being $(-1)^n$.

² Consider that $\sum_{i=0}^{\infty} x_i^3 = \sum_{i=0}^{\infty} \underbrace{(x_i)}_{\text{converges}} \underbrace{(x_i^2)}_{\rightarrow 0, +}$ but we still have a counterexample!

so as long as $\alpha_0 - \beta_0$ are chosen intelligently here and for every subsequent quadruplet of terms, if their series converges then we can conclude. Let's assume they were chosen well and consider now the sum of cubes.

$$\begin{aligned} x_0^3 + x_1^3 + x_2^3 + x_3^3 &= y_0^3 - (y_0 - \alpha_0)^3 + y_2^3 - (y_2 + \beta_0)^3 \\ &= 3\alpha_0 y_0^2 - 3\alpha_0^2 y_1 + 3\alpha_0^3 - 3\beta_0 y_2^2 - 3\beta_0^2 y_2 - 3\beta_0^3 \end{aligned}$$

Now, taking $y_0 = y_2$ and $\alpha_0 = \beta_0$ (what's stopping us? As long as we reduce their value for the next quadruplet of terms, the sum of x_i will still conclude) this simplifies to

$$x_0^3 + x_1^3 + x_2^3 + x_3^3 = -6\alpha_0^2 y_0$$

but our only condition on α_0 is that it should decrease (so if we say more generally that $\alpha_i = y_{4i} - y_{4i+1} = y_{4i+3} - y_{4i+2}$, then $\alpha_i \rightarrow 0$ is sufficient) and the same is true for the x_i . So we can make a straightforward counterexample - we just want the sum of $-6\alpha^2 x$ to diverge. So let's just make it so $\alpha^2 x$ is the harmonic sequence:

$$\{x_i\}_{i=0}^{\infty} = \begin{cases} x_{4n} = \frac{1}{\sqrt[3]{n}} \\ x_{4n+1} = -\frac{1}{\sqrt[3]{n}} - \frac{1}{\sqrt[3]{n}} \\ x_{4n+2} = \frac{1}{\sqrt[3]{n}} \\ x_{4n+3} = -\frac{1}{\sqrt[3]{n}} + \frac{1}{\sqrt[3]{n}} \end{cases}$$

Constructed in this way we have that $|x_i| \rightarrow 0$ and $x_{4n} + x_{4n+1} + x_{4n+2} + x_{4n+3} = 0$, meaning $\sum_{i=0}^{\infty} x_i = 0$, but $x_{4n}^3 + x_{4n+1}^3 + x_{4n+2}^3 + x_{4n+3}^3 = -\frac{6}{n}$ meaning the latter diverges as desired!

One final note: look closer at the values of x_i we found. You will notice they turn out to be the same as what you saw in the seminar... but with an additional 0 thrown in! So whilst at the time you were told that there was no strong motivation for creating this sequence and it was partially down to intuition, now you have a strong motivation for it! Every counterexample, no matter how bizarre, has a reasoning behind it³.

So what does this illustrate? Initially, you would be persuaded that the answer was indeed true, and that the series of cubes would also converge. Then, by eliminating possible cases (it works if the sequence is positive, if it alternates and goes to 0 monotonically...) you end up considering a new case (alternating but *not* going to 0 monotonically) which gives you the counterexample.

Problem. Let $\{x_i\}_{i=0}^{\infty}$ be a sequence of real numbers such that $\sum_{i=0}^{\infty} x_i$ converges, and $\{y_i\}_{i=0}^{\infty}$ be a sequence of positive real numbers such that y_i converges to 0. Then $\sum_{i=0}^{\infty} x_i y_i$ converges. True or false?

Answer: No, duh. Just look above.

1.3 Trust your heart, not your head

True/false questions are a test of intuition and skill; more than anything, they test how fast you can chase down the correct answer between true and false, after which the proof should not be too difficult to formulate. In this sense they often a deceitful foe, always trying to mislead you by leading you down a false alleyway. You might see a statement and think it to be obviously true, and find the ensuring counterexample leaves you traumatised, shaken and unable to trust your gut feeling any more. This is the most dangerous thing you can do to yourself: if you are too unsure whether something is true or false then you will never have the courage to try and do the question, out of the perpetual fear you are heading towards the wrong answer. Instead, the key is to find the balance between confidence and being headstrong - trust your gut feeling, but always be prepared for the possibility that you

³ I lied. I have no idea what motivated John Conway to create the Conway Base-13 function.

may have not been correct. It happens to the best of us - the key is being able to realise quickly and without losing focus.

2 THE ART OF FINDING A COUNTEREXAMPLE

So let's say the queen of analysis⁴ - intuition - has told you the way to go is "false". So how does one find a counterexample? Let's look at a few interesting lines of approach, and some common pitfalls to avoid.

2.1 Negation goes brrrrr

As mentioned before, at its core a counterexample is just a creature born from mathematical logic that should be " $\mathbb{A}$ but not $\mathbb{B}$ ". So what exactly *is* "not $\mathbb{B}$ "?

Consider for example $\mathbb{B} :=$ "the limit as $n \rightarrow \infty$ is $+\infty$ ", describing a positive real sequence. What, then, is a sequence that does not go to infinity? If I were to take a microphone and head to the street and start asking random strangers, they probably wouldn't understand the question. If I were to do this at EPFL with first-year students, there is a fair chance most of you would reply "bounded!". This makes sense: the sequence isn't going to infinity, so it isn't going to become arbitrarily big, so it should be bounded, right? Let's take a step back and bring out our mathematical logic toolbox.

Firstly, recall the definition of "the limit is infinity". A sequence of (positive) reals goes to infinity if

$$\forall N, \exists M : n > M \implies x_n > N.$$

Let's break this down into logical parts. We need, for all numbers N , the existence of some other number M - dependent on N - such that after the M^{th} term, the sequence of x_i is always larger than N . Visually speaking, if you picture the sequence as a graph of disconnected points - since, after all, it is a function from $\mathbb{N}$ to $\mathbb{R}$ - then for any horizontal line you draw, the graph will eventually go above that line and **stay above it**. It can have spent time above the line before this moment, but after this point it stays above the line permanently. In particular, this is clearly a stronger condition than just being unbounded.

What, then, is the negation of this statement? Firstly, here is a very hand-wavey principle:

- The logical negation of $\forall$ is $\exists$, and vice versa.

So if you want to negate a statement, you just have to switch all $\forall$ to $\exists$, and turn $\implies$ into $\not\implies$ (and vice versa). So in other words, the negation of the previous statement would just be

$$\exists N : \forall M, n > M \not\implies x_n > N.$$

Let us convince ourselves this is indeed the negation by thinking it through step by step. Our initial statement said for any N , we would eventually surpass it permanently. If this is not true for all N , then there needs to be some N where it is not necessarily true (but it doesn't need to be all N). Not being surpassed permanently means for every choice of a M , the condition that "the sequence is permanently larger than N " fails to hold. Visually speaking, the graph will return below some horizontal lines arbitrarily many times. Indeed, we can use our understanding to formulate another negation that is somewhat easier to understand:

$$\exists N : \forall M, \exists n > M : x_n < N.$$

This is clearly not the same as being bounded: being bounded is a much stronger condition, which is actually equivalent to having $\forall n > M$ rather than $\exists n > M$ in the previous formulation.

The reason why I call this principle hand-wavey is because just swapping these two symbols is not an exact art; you should not just blindly exchange them (whilst

⁴ If intuition is the queen, then what is the king? Keep reading to find out.

not forgetting to negate all the other symbols!) without truly understanding what the negation represents.

So why is this useful? No explanation is more illuminating than a problem:

Problem. Consider a sequence of positive reals $\{x_i\}_{i=0}^{\infty}$ such that $\sum_{i=0}^{\infty} \frac{x_i}{i^2+1} = +\infty$. Then $\lim_{i \rightarrow \infty} x_i = +\infty$. True or false?

To see the usefulness of proper negation, we consider the approaches taken by two fictional students, A and B. A thinks $\pi = 3$ and should clearly be in GM⁵ BA 1. B recently read the bestselling *The Art of War: How to solve true/false questions in Real Analysis*TM and is now an expert on negating stuff.

A reads the question, and after they are done crying, begins to look for a counterexample. Now, we want a sequence satisfying the condition, but that doesn't go to $+\infty$. Now, if a sequence is bounded it clearly doesn't go to $+\infty$, so A reckons he can just consider a bounded sequence. If the sequence is bounded, say by M , then we have $\sum_{i=0}^{\infty} \frac{x_i}{i^2+1} \leq \sum_{i=0}^{\infty} \frac{M}{i^2+1} \leq \frac{M\pi^2}{6}$ (the second inequality is not that necessary; what is more important is the sum of the square reciprocals is convergent). A thinks he has therefore proved the contrapositive, writes "QED" in enormous letters, and is now a low-ranking member of the St. Petersburg mafia.

Meanwhile, B recalls what she has learnt, and correctly negates the statement. She knows that not going to infinity just means the function returns below a certain threshold indefinitely. But with this in mind, it is very easy to manipulate your values in order to have $\sum_{i=0}^{\infty} \frac{x_i}{i^2+1}$ diverge. You only need some of your terms to be big, and the rest can be small. For example, taking a sequence like 10, 0, 100, 0, 1000, ... will clearly work. B is done with the exercise after about 30 seconds and goes on to win the Fields medal the following year.

Therein lies the usefulness of negating properly: it ensures that you do not incorrectly consider hypotheses that are too strong or too weak, and that you know exactly what you are looking for in a counterexample.

2.2 Milking examples to give counterexamples

Often, vital clues on the correct counterexample can be found in an example; alternatively, sometimes all it takes is a little tweak to an example to get the behaviour you want. For example, consider the following question:

Problem. Consider a sequence of positive reals $\{x_i\}_{i=0}^{\infty}$ such that $\lim_{i \rightarrow \infty} x_i = 0$. Then $\exists n \in \mathbb{N}$ such that $\sum_{i=0}^{\infty} x_i^n < \infty$. True or false?

In other words, can we raise every sequence converging to 0 to a high enough power to ensure that the corresponding series will converge? There is no confusion here over what exactly we are looking for (the negation of not going to infinity is just going to infinity, since the series in question only has positive values). So, in order to begin our hunt for a counterexample, let's consider some examples.

Obviously, let's try and take a sequence that doesn't already have a convergent sum, and then see what power we have to raise it to, to obtain a convergent series. The classical example is the harmonic series; here, taking $n = 2$ is already sufficient. If we consider a sequence that goes towards 0 less rapidly, for example $\frac{1}{\sqrt[k]{i}}$, then taking $n = k + 1$ is sufficient. We observe this behaviour where once the sequence goes to 0 faster than the harmonic sequence, then its series will converge. In other words, we want a sequence where no matter what power you raise it to, it does not "overtake" the harmonic sequence. Alternatively, if we consider now the sequence of $\frac{1}{x_i}$ then the condition is that they must go to ∞ , but the question is whether after applying a high enough power they will always overtake the integers.

⁵ I have many friends who are mechanical engineers. You can't sue me. I am the law. I am the one who knocks.

Clearly, any kind of power function is not going to work; even if you take $i^{0.00001}$ then $n = 10001$ would suffice. No; what we need is something *real* slow. Ideas like $\sqrt[n]{n}$ are funky and all, but it's fairly difficult to assess whether they tend to ∞ or not. At this point, you just need to revert to what you know about the growth rate of classic functions: in particular, $\ln(x)$ grows slower than any power function and so $\frac{1}{\ln(x)}$ works because no matter what power n you take it to, $\ln(x)^n$ always grows slower than x . Don't forget that when you provide a counterexample, you have to justify why it works - but proving that this counterexample works is left to the reader, so enjoy.

Another example is a function that is discontinuous but integrable everywhere. The classic example of a function that is discontinuous everywhere is the indicator function of $\mathbb{Q}$. Unfortunately, this function isn't integrable, but the idea of defining a function as 0 on the irrationals and nonzero elsewhere is quite appealing. The key is that the function should take a lot of arbitrarily small values, so that the upper Riemann sum can be as small as we want, depending on interval width. Armed with this, there are a number of functions that you can find, but again, this is left to the reader as fun exercise (if you want a hint: since you want the values to be small, the function should ideally take into account the denominator [of the number written as a irreducible fraction], since this will usually be quite big, but perhaps not the numerator).

2.3 Famous things are famous for a reason

Our beloved student A (he of the organized-crime fame) missed one enormous clue as to why bounded was not the negation of going to infinity. If every sequence that was unbounded went to infinity, then why would we bother introducing the latter definition? For example, we just say that sequences that don't converge are divergent. We don't introduce a new definition, because we can just say divergent = *not* convergent. Similarly, decreasing isn't defined as *not* increasing for a reason - it's not the negation! You can obviously have sequences that are neither increasing nor decreasing. More generally, if we went to great pains to give you a new definition, then you should be wary of true/false questions that suddenly present a much simpler set of properties for that definition, or a property that you have not seen before. If all integrable functions were continuous, then surely you would have been told a fact this important already.

One of the best things you can do is have a list of classic counterexamples in mind. The best reference for this kind of thing is Gelbaum and Olmsted's *Counterexamples in Analysis*, which at the time of writing could be downloaded at https://faculty.ksu.edu.sa/sites/default/files/_olmsted_1.pdf. Otherwise, when you are doing practice problems, and you encounter a particularly remarkable counterexample, keep it in mind - it may be useful someday!

3 PROVING 101 FOR DUMMIES

You've gone on a hunt and there are no counterexamples to be seen, so the time has come to make a proof. However, let's say you don't see any obvious way to proceed, and to show that the properties just directly imply the conclusion. So how does one go about constructing a proof, exactly? And how do you make sure your proof doesn't have any holes in it?

3.1 How to write a nice proof

Surprisingly, proofwriting is a skill that many people neglect, despite how critical it will be in your mathematical journey. Let's quickly review the fundamentals of writing a nice proof.

- Clarity is key. Adopt a nice structure - state intermediary results as lemmas, then prove them, and put a nice QED at the end of them to notify the reader you have shown the result and are moving on. Then, later refer to these lemmas by name in order to prove your main point. Breaking down a laborious proof into easily digestible subparts will both ensure that you can more easily verify you are not making mistakes, and will be a blessing for the poor soul that has to correct your work.
- Reread your work carefully! A small typo can have a big effect in a numerical problem, and it is very easy to convince yourself that your fallacious argument is correct. A simple way to check is to just consider extremal cases and see if the argument holds throughout. For example, consider the following correct statement with an erroneous justification:
An alternating sequence whose absolute value goes to 0, has a convergent infinite sum. This is because both the positive and negative parts converge, and so we can evaluate them separately and then add their respective sums.
 This is of course true: the alternating harmonic series would be an easy counterexample. Check your claims, equalities and inequalities against simple examples and do sanity checks where necessary.
- Rigour is important, but be sensible when applying it. You do not need to re-prove in detail that $\sqrt{2}$ is irrational, but do not grandly declare that it is obvious that the roots of some obscure polynomial you cooked up are irrational either. Even a little justification goes a long way.
- One clean proof is better than two half-baked ones. If you are low on time, make sure you ensure what you have already finished is correct rather than trying a wild bluff to solve the hardest question on the exam.
- Your teachers are human. If your proof is 4 pages long, invokes algebraic geometry and is cobbled together with duct tape and hopes and dreams, it's probably not the intended solution. Always look for nice and simple ideas before trying to invent your own number system in order to solve a problem. Also, if you use some extremely strong result, you run the risk of using circular reasoning by accident. For example, if I were to ask you why the n th root of 2 was irrational, you could justify using Fermat's Last Theorem⁶, but I would be amazed if you could explain to me why this result does not itself depend on the cube root of non-cubes being irrational.

You have all seen by now the aforementioned fact that $\sqrt{2}$ is irrational. Let's now consider a good proof that $\sqrt{n}$ is rational if and only if $\sqrt{n}$ is an integer.

⁶ Otherwise, if $2 = (p/q)^n$, we have $q^n + q^n = p^n$.

Proof. Suppose for the sake of contradiction that there exists some n such that $\sqrt{n}$ is rational but not an integer.

Lemma. *Under the above assumption, there exists some squarefree m such that $\sqrt{m}$ is rational but not an integer.*

Proof. Let k^2 be the largest square number dividing n . We prove that $m := n/k^2$ suffices. Note firstly that $\sqrt{m} = \sqrt{n}/k$ is rational. Suppose now that m is not squarefree, and let ℓ^2 be a square dividing m . It stands to reason that $(\ell k)^2$ is a larger square number dividing n , contradicting the maximality assumption. $\square$

We will now work with m , which we can write in the form $p_1 \cdot p_2 \cdot \dots \cdot p_r$ as a product of distinct primes. Let $\sqrt{m}$ in irreducible form be x/y . We then have

$$x^2 = y^2(p_1 \cdot p_2 \cdot \dots \cdot p_r).$$

It follows that x^2 (and also x) is divisible by all the p_i , meaning x is divisible by m , and the RHS is divisible by m^2 . Dividing both sides of the above equation by $p_1 \cdot p_2 \cdot \dots \cdot p_r$, we get that the RHS is still divisible by m , and so m divides y^2 . But by a similar argument, m divides y , contradicting the irreducibility of x/y . $\square$

Now consider a different proof of the same statement.

Proof. Take an n that has a square root that is rational but not an integer. Write the prime factorisation of n and if there are any squares we can remove them (as this divides the square root of n by an integer, which preserves its rationality) so we are left with something where each exponent is at most one - a product of primes. Call this n instead. Now for every prime going into n we know that the square root of that prime is irrational. To see that the square root of a prime is irrational, suppose otherwise and let $\sqrt{p} = p/q$. Then p^2 and q^2 are both integers, and since p^2 is q^2 times a prime, we know the prime divides p^2 and we note that if a prime divides x^2 the the prime must divide x , because writing the prime factorisation of x as $p_1^{q_1} p_2^{q_2} \dots$ then the prime factorisation of x^2 is $p_1^{2q_1} p_2^{2q_2} \dots$ and one of the p_i is actually p but this means that as $2q_i$ is an integer then q_i is also an integer and it was already there. Therefore the square root of n is the product of distinct irrationals and itself irrational. $\square$

Do you see how much worse the second proof is? Firstly, writing all your ideas in a rush one after the other makes the structure impossible to follow. Secondly, take the time to label your variables appropriately - the statement $\sqrt{p} = p/q$ where you introduce a second instance of p referring to a new prime is a crime against mathematics, as is replacing n with a new n rather than just using a new letter. If the Romans left us with 26 letters, it was not so that you could abuse the same two constantly. Finally, the justification doesn't even work - the product of distinct irrationals is not itself irrational!

3.2 The two contratom bombs: proof by contrapositive and contradiction

If you don't see any immediate proofs jump out at you, then two routes that almost always work at to either prove the contrapositive, or proceed with a proof by contradiction. Let's dissect both of these two methods and understand how they work.

Firstly, the proof by contrapositive. Again, let us lean on some mathematical logic: one property of logical implications is as follows:

$$(A \implies B) \equiv (\neg B \implies \neg A).$$

For example, if I were to ask you to prove that any function that isn't integrable can't be continuous everywhere, it suffices to point out that every continuous function is integrable. A proof by countrapositive is still a proof, though - you are

just taking the metro in the opposite direction, but you still have to formally prove something.

A proof by contradiction is a sort of generalisation of a proof by contrapositive. Instead of having to do any actual proving, you just assume that what you want isn't true, and show that this implies something is fundamentally wrong with the universe. Expressing this in a formalised manner with mathematical logic like above, we can state:

$$\exists x \in (\mathbb{A} \cap \neg \mathbb{B}) \implies \text{Loud explosions and screaming noises.}$$

The fun thing is that you don't really have a set route, and there are usually dozens of ways to finish now. You just have to identify in what the existence of something which is both $\mathbb{A}$ and not $\mathbb{B}$ creates an impossibility. The best thing is that usually you can pin down exactly what it is that is troubling about the aforementioned assumption, and then you just have to describe it mathematically, and you're done. Let's look at an example.

Problem. *Every monotone function has a countable number of discontinuities. True or false?*

Here, you recognise that a discontinuity is just a point where the function makes a little jump upwards, due to the supposed monotonicity. Intuitively, this appears to be true. No matter how hard we look for a counterexample, we just can't seem to create a function with an uncountable number of discontinuities - countably infinite is certainly possible (easy example: $f(x) = \lceil x \rceil$) but transitioning from countable to uncountable just doesn't seem plausible. However, if you just assume a function is monotone, it doesn't naturally ensue by the definition or anything that the number of discontinuities ought to be countable. Let's try a proof by contradiction.

Suppose now we have a function which is monotone and has an uncountable number of discontinuities. What seems to be the issue now? For a start, there ought to be some place with a lot of these discontinuities - if they were all spread out "thinly" enough - in the sense we only ever found a countable number of them on an interval - then it seems implausible that we would have an uncountable number over all of $\mathbb{R}$. Now, if there were an uncountable number of discontinuities on a finite interval then the issue is that the function seems to increase way too rapidly - no matter how small the jumps are, if there are an uncountable number of them then their "sum"⁷ appears to be infinite, no matter how small they are. So in other words, the contradiction appears to be with the fact the function should be bounded on this supposed finite interval, where there are way too many discontinuities.

Now, of course, comes the hard part - converting our blah-blah worthy of someone studying in any other degree into proper maths. Firstly, let's prove there is indeed some interval with an uncountable number of discontinuities. To do this, we can again proceed by contradiction - suppose now that in every interval of the form $[z, z + 1[$ we have a countable number of discontinuities. Since we clearly have that $\mathbb{R} = \bigcup_{z \in \mathbb{Z}} [z, z + 1[$, it follows that the total set of discontinuities is a countable union of countable sets, or in other words (by one of the very first exercises you encountered in Algebraic structures) countable! This clearly contradicts the fact it was uncountable by assumption, so we know there is some interval $[a, a + 1[$ - or, extending it a little, $[a, a + 1]$ - with an uncountable number of discontinuities; let us refer to them as $S \in [a, a + 1]$

We know that we want to prove the sum of the discontinuities is infinite, as we have that $f(a + 1) - f(a) > \sum_{x \in S} (\lim_{y \rightarrow x^+} f(y) - \lim_{y \rightarrow x^-} f(y))$, which is the

⁷ A semantic interlude: Of course, an infinite sum is not actually a sum but rather an abuse of notation that refers to the limit of the finite sums as you add more and more elements. In this sense, there is no proper way to define an infinite uncountable sum, because the very nature of a limit is that you need countable steps, whereas in the uncountable case you can't just add in numbers one by one. However, for our purposes here it suffices to just say that if an uncountable set has finite sum we mean that all its countable subsets have finite sum, and if it has infinite sum we mean that there is at least one countable subset with infinite sum.

formal way of saying the overall increase bounds the sum of all the jumps. Now, all we know about these jumps is that they are positive reals, and with just this information we should be able to conclude.

Again, some blah-blah-wording-thinking-reflecting is needed here. Of course, the jumps can be arbitrarily small, so the critical factor here ought to be the fact that there are an uncountable number of them. One way to prove that a sum of positive reals is infinite is to show that infinitely many of them are greater than some $\varepsilon > 0$. Now, for a countable set this needn't be true - for example, the harmonic series has an infinite sum, but there are only finitely many reciprocals greater than any given ε - but because we now have an uncountable set, it seems natural that because there are *so many* of them we would end up with too much above a given point somewhere, right?

Before we let the spirit of a physicist take a hold over our soul, let us stop and go back to formal mathematics. How can we formalise the notion of there being too many above a given point? We can take inspiration of our proof that there were uncountable many discontinuities in an interval.

Consider now the intervals $[1, \infty[, [\frac{1}{2}, 1[, [\frac{1}{3}, \frac{1}{2}[, \dots, [(n+1)^{-1}, n^{-1}[, \dots$ and put each jump size into the interval containing it. Now note that by the same logic as above, they can't all contain a finite (and therefore countable) number of jump sizes, so there is some interval with an (uncountably) infinite number of jump sizes. In particular, this means that their total sum is infinite - which contradicts the fact that the sum must be bounded by $f(b) - f(a)$! And we are done.

What are the key steps here? By imagining an object with the properties but not the conclusion, you can identify a problematic characteristic - for example, here the fact that the function seemed to go to infinity before some finite point, which is of course not allowed for a monotone function, as how would you define it from that point onwards? Then, you must formalise and understand what the mathematical interpretation of your issue is (so here, being unbounded on a finite interval) and then use what you know about the object to show the issue holds, and so we have a contradiction. Of course, there is still some proving to be done here, but not only are we more flexible (there are usually many different contradictions you can arrive at, and many routes to take) but you get to assume more than before - rather than just assuming $\mathbb{A}$ in order to show something, you get to assume $\mathbb{A}$ and $\neg\mathbb{B}$, give you way more toys to play with. This is why a proof by contradiction is the king of all analytic techniques - it is powerful, failsafe (after all, any proof can be turned into a proof by contradiction by tweaking it, so a proof by contradiction will always exist), easy and does not usually require a brilliant idea, but rather just the ability to formalise your observations.

3.3 Using all the ingredients to make your cake

Another important factor you should always keep in mind is if you used all your properties in order to prove your conclusion is true. Mathematicians have a very potent weakness - they want to ask questions that are as *strong* as possible, in the sense that you don't use any extraneous information. Why would you bother using more stuff than you need, just to obtain a result that's weaker than it needs to be? To illustrate what we mean, consider the following example.

Problem. Let $f(x)$ be a function satisfying $f(x) + f(y) = x + y$ for all $x, y \in \mathbb{R}$, that is continuous at $f(2021)$. Then $f(x) = cx$. True or false?

Consider the following solution:

For a start, we have that $2f(1) = f(1) + f(1) = f(2)$, and more generally $nf(1) = f(n)$ by induction. Furthermore, since $f(0) = f(0 + 0) = 2f(0)$, we have that $f(0) = 0$. This means that $f(n) + f(-n) = 0$, so $f(-n) = -nf(1)$. So we have successfully proved our desired result on the integers.

Furthermore, we have by a generalisation of the earlier steps, using $k \in \mathbb{R}$ instead of 1, that $f(nk) = nf(k)$. By swapping n and k we now obtain $f(kn) = kf(n)$. It follows that $kf(1) = f(1k) = kf(1)$ as desired, taking $1 \in \mathbb{N}$. As k is arbitrary, we are done.

Do you see the issue in this solution? (Hint: the swap is somewhat illegal). In any case, an immediate red flag should have been the fact that one of the hypotheses is untouched - if you didn't need to use the continuity of f somewhere, then why on earth would the problem state it as a hypothesis? If your proof does not need all the hypotheses stated, then this is very suspicious.

Two cautionary notes: firstly, *don't* take this idea to its logical extreme. Just because the previous problem used continuity at $f(2021)$ does *not* mean the value 2021 is in some way important to your proof; logically, if it was stated with 2020 instead, it shouldn't change a thing - so don't get stuck on wondering why the question states 2021 in particular. Another thing you should be careful is mistakenly believing you didn't use a hypothesis when you actually did, and being suspicious of a completely valid proof as a result - it is very easy to implicitly use a hypothesis without realising. For example, if one of the hypotheses is that $f(x)$ is injective, and at some point you write $(x - y)(f(x) - f(y)) = 0 \implies (x - y) = 0 \implies x = y$, then you implicitly used the fact that the case $f(x) - f(y) = 0$ also leads to $x = y$ by injectivity. So before assuming you forgot to use a hypothesis and your proof must be flawed, make sure you actually forgot it! Speaking of using hypotheses without realising...

3.4 Fallacies and how to avoid them

The easiest way to make a mistake in a proof is to falsely assume something without realising you have done so. This kind of error can be extremely subtle.

For example, say I gave you a set of positive reals. It would be a mathematical fallacy to consider the smallest element of this set, since it doesn't necessarily exist - as you all know, the infimum is not necessarily a minimum. This might seem obvious, but it is quite a common error in practice to just use the smallest element during a proof without distinguishing for the case where there is no smallest element.

We can make this even more subtle - now consider a periodic function. It seems perfectly natural to consider the smallest period of this function, because visually we see period functions as those which do something interesting and then return to where they were to repeat, and we intuitively have a notion of there being a *first* time they return - for example, $\sin(x)$ is of course 10π -periodic, but its "fundamental"⁸ period is 2π . Nevertheless, there are periodic (non-constant) functions with no smallest period. The simplest example is our good old friend the indicator function of $\mathbb{Q}$ - clearly, every rational is a period of this function, but there is no *smallest* period per se.

Another classic example is, when working with a function f , to say something like "consider x, y such that $f(x) \neq f(y)$, and...". How do you know these two points exist? Of course, they are nearly always there - but if the function is constant, they wouldn't, so you can't assume they exist - at the very least, you have to treat the constant case separately, however trivial it may be.

There are dozens of such minute errors you can run into - forgetting the negative case when taking a square root, assuming a function that goes to infinity will at some point be equal to a large number (as opposed to greater than), assuming a countable set may be ordered countably (this one is particularly dangerous - all the time, people have a countable set S and then start talking about $s \in S$ and then "the next biggest element of S after s ", which of course does not need to exist - it's just a corollary of the first fallacy we discussed), or even just forgetting the classic case when something is equal to 0.

⁸ Trust a physicist to give a definition for something that doesn't necessarily exist. Yes, I know the fundamental period is primarily defined for sine curves. Let me nitpick in peace.

There is no easy way to avoid making an error like this - the best way to do so is to understand mathematical structures well (a set needn't have a smallest element, a function needn't be bounded). Another thing you can do is have a couple of edge cases in mind - by which I mean subjects that really push the boundary of what the properties entail; for example, the indicator function of $\mathbb{Q}$ when talking about periodic functions, or Conway's base-13 function⁹ when talking about surjective functions - and checking at every point of your proof if you made an assumption that wouldn't be true for the edge cases. Did I assume there's a smallest period? That's not true for the rational indicator function, so I should handle the case of "no shortest period" separately. Did I assume there was an interval where my function was bounded? Conway's base-13 function doesn't behave that way, so I need to accommodate for this in my proof. Two steps forward, one step backward.

⁹ https://en.wikipedia.org/wiki/Conway_base_13_function

CONCLUSION

True and false questions really are a battle between the student and the teacher - the questioner will always be looking to trick you, to have a sneaky counterexample or a result that seems completely counterintuitive but true. You shouldn't live in complete paranoia - you will never get an exam containing only impossible-to-find counterexamples, as that would just be a pedagogical failure and a nightmare to correct - so don't doubt your instincts. Your best tool is your own intuition, and the only way to hone that intuition is through practice, practice and more practice.

An important thing to keep in mind - somewhat unrelated to how you should tackle the questions themselves - is how low the barrier for success was the last time an exam like this was held. If your final exam turns out to have 19 problems on it, it would be unreasonable to expect anyone to finish them all within the time limit, so go for your favourite topic first and play to your strengths. Remember that a full solution is always better than two partial ones, so try and make it to the end of a problem before moving on, rather than giving up halfway through a proof. One piece of advice that is quite pertinent but that I find absolutely useless is "don't stress" - it's clearly what you should do, but it's just about impossible not to.

Last (and certainly not least) you should keep in mind that you chose to study mathematics because you enjoy it¹⁰ and it shouldn't be something that causes you constant anxiety and worry¹¹. Remember to try and enjoy the questions, and see the exam (and this goes for all your exams) as an opportunity to prove what you have learnt this semester and what you are capable of, rather than a do-or-die survival test. Take pride in your work, and always indulge in elegant proofs and beautiful counterexamples. I hope to see you in BA2.

Merry Christmas!

¹⁰ Sadly, half of you probably don't love it and are just here to sell out to Big Finance. I'm judging you.

¹¹ I know it does, I know. Stop crying.